Hibiscus sabdariffa L. extracts inhibit the mutagenicity in microsuspension assay and the proliferation of HeLa cells.

Hibiscus sabdariffa L. is used as a refreshing beverage and as a traditional medicine. The objective of this study was to determine the in vitro effect of phenolic compounds present in aqueous, ethyl acetate, and chloroform extracts of H. sabdariffa against mutagenicity of 1-nitropyrene (1-NP), and also the antiproliferative effect of these extracts. Inhibition of cell proliferation and DNA fragmentation were tested on transformed human HeLa cells. The hot aqueous extract (HAE) contained 22.27 +/- 2.52 mg of protocatechuic acid (PCA) per gram of lyophilized dried extract, and was not statistically different from the cold aqueous or chloroform extracts; the ethyl acetate extract produced the least amount of PCA. The H. sabdariffa extracts inhibited mutagenicity of 1-NP in a dose-response manner. The inhibition rate on HeLa cells of HAE was also dose-dependent. The HAE did not induce DNA fragmentation. The results suggest that H. sabdariffa L. extracts have antimutagenic activity against 1-NP and decrease the proliferation of HeLa cells, probably due to phenolic acid composition.